

Practice Quiz 02D — MATH 2055 — Fall 2026 — Solutions

Practice for Quiz 02

Conditionals, contrapositives, and free and bound variables.

1. Let R mean *the robot moves* and E mean *the emergency stop is off*. The statement is $R \rightarrow E$.
- (a) Write its converse and contrapositive as formulas. Make one four-row truth table and say which is equivalent to $R \rightarrow E$.

Solution The converse is $E \rightarrow R$ and the contrapositive is $\neg E \rightarrow \neg R$.

R	E	$R \rightarrow E$	$E \rightarrow R$	$\neg E \rightarrow \neg R$
F	F	T	T	T
F	T	T	F	T
T	F	F	T	F
T	T	T	T	T

The contrapositive is equivalent to the original conditional; the converse is not. ■

- (b) In $\exists z \in \mathbf{Z} \forall w \in \mathbf{Z} (z + w = d)$, list the free variables and the bound variables.

Solution The variables z and w are bound by their quantifiers. The variable d is free. ■